

ISO 19115 / DCAT Compliance Gap Analysis

AK Region mdEditor Profiles & Schemas — baseline assessment (pre-v1.1.0)

Status note: This document reflects the original gap analysis performed against schema/profile versions `ak-proj-schema-v1.0.0`, `ak-prod-schema-v1.0.1`, `ak-prod-profile-v1.0.2`, and `ak-proj-profile-v1.0.1`. Most items below were addressed in the **v1.1.0–v1.4.0** releases — see the [README](#) for what changed and the **Status** column in each table here for current disposition. This doc is retained as the historical record of *why* the current schemas look the way they do, including two reversals: `accessLevel`/`bureauCode`/`programCode` were added in v1.1.0 and removed in v1.2.0, and `metadataInfo.metadataStandard` was added in v1.1.0 and removed in v1.3.0/v1.4.0 — both reversals happened once it was confirmed the fields have no corresponding path in mdEditor's actual mdJSON schema.

Context

These files are mdEditor **JSON Schema validators** (`ak-*-schema-*.json`) and **profile/UI configs** (`ak-*-profile-*.json`). The underlying data model (mdJSON) already implements ISO 19115-1/19115-2, so the field vocabulary is ISO-native. The gaps identified here are about which *ISO-mandatory* elements the validators failed to require, plus DCAT properties that have no ISO analog and were not enforced at all.

1. Gaps vs. ISO 19115-1 Core Elements

ISO 19115-1 Table 1 (Annex A) defines a "core" set of elements that are **mandatory (M)** or **conditional (C)** regardless of profile.

ISO Core Element	Obligation	Baseline status	Status as of v1.1.0
Dataset title	M	✅ Enforced ((<code>citation.title</code>) required)	✅ No change needed
Dataset reference date	M	✅ Enforced ((<code>citation.date</code>) required)	✅ No change needed
Metadata point of contact	M	✅ Enforced ((<code>metadataContact</code>) requires (<code>author</code>) + (<code>pointOfContact</code>))	✅ No change needed
Metadata date stamp	M	✅ Enforced ((<code>metadataDate</code>) requires (<code>creation</code>) type)	✅ No change needed
Abstract	M	✅ Enforced ((<code>abstract</code>) required)	✅ No change needed
Dataset language	M	❌ Not present anywhere in (<code>resourceInfo</code>) required list	✅ Fixed — (<code>language</code>) added as required in both schemas
Dataset topic category	M	❌ Not present — no (<code>MD_TopicCategoryCode</code>) requirement	✅ Fixed — (<code>topicCategory</code>) added as required in both schemas
Geographic extent (if data has spatial component)	C	⚠️ (<code>extent.geographic.containsData</code>) explicitly disabled ((<code>false</code>)) in both profiles; never required in either schema	✅ Fixed — required in schema (conditionally for products, unconditionally for projects); un-hidden in both profiles
Reference system (CRS)	O, effectively required for usable geospatial records	❌ No (<code>referenceSystemInfo</code>) requirement in any schema; not shown in UI profiles	✅ Fixed in schema (both). ⚠️ Profile field path unverified — see README Known Limitations
Spatial representation type	O, expected for spatial data	❌ Absent	⏸️ Not addressed — carried forward as a future improvement, not blocking for v1.1.0
Lineage	C (required when data quality info isn't otherwise supplied)	❌ Entirely hidden/disabled in product profiles, not required in product/project schema	✅ Fixed for products — (<code>lineage.statement</code>) required. Projects have no lineage tab in mdEditor's nav, so not applicable there.
Metadata standard name/version	O, but expected when claiming ISO conformance	❌ Not populated or enforced	❌ Removed from scope. Added in v1.1.0 as (<code>metadataInfo.metadataStandard</code>), then removed once no confirmed field path could be found in mdEditor's mdJSON schema. Unlike the POD fields below, this is a legitimate ISO element — reintroduce it if/when the correct mdJSON path is confirmed (see README Known Limitations).
Distribution format	O, but needed for usable/DCAT-mappable records	❌ (<code>distributionFormat</code>) explicitly hidden in every profile	✅ Fixed — required in schema; un-hidden in both profiles

Baseline conclusion: the schema enforced the administrative/citation backbone well (contacts, dates, title, keywords) but skipped almost everything that makes a record *geospatially* ISO-compliant — topic category, language, extent, CRS, and lineage.

2. Gaps vs. DCAT / DCAT-US (Project Open Data)

Since AK Region is a federal program, the relevant compliance target for public catalog submission is **DCAT-US / POD Schema** (what powers Data.gov).

DCAT-US / POD Property	Maps from	Baseline status	Status as of v1.1.0
<code>dct:title</code>	<code>citation.title</code>	✓	✓ No change needed
<code>dct:description</code>	<code>abstract</code>	✓	✓ No change needed
<code>dcat:keyword</code>	<code>keyword</code>	✓	✓ No change needed
<code>dct:modified</code>	<code>citation.date</code>	✓ (type not constrained)	⚠ Not addressed — date-type constraint for "modified" specifically remains a future improvement
<code>dcat:contactPoint</code>	<code>pointOfContact</code>	✓	✓ No change needed
<code>dct:identifier</code>	<code>citation.identifier</code>	⚠ Only required in project schema, not enforced as unique/persistent	✓ Fixed — required (<code>minItems: 1</code>) in both schemas
<code>dct:publisher</code>	<code>citation.responsibleParty</code> (role=publisher)	⚠ Required in product schema only — missing from project schema entirely	✓ Fixed — <code>publisher</code> role now required in project schema too
<code>dcat:distribution</code> → format/mediaType	<code>transferOption.distributionFormat</code>	✗ Hidden in every UI profile; never required in schema	✓ Fixed — required in schema; un-hidden in both profiles. Project schema also gained a <code>resourceDistribution</code> requirement it previously lacked entirely.
<code>dct:license</code> / <code>dct:accessRights</code>	<code>constraints</code>	✗ No <code>constraints</code> requirement anywhere	✓ Fixed — new required <code>constraints</code> section (license/useConstraints/legalConstraints) in both schemas; project profile also gained a missing "Constraints" nav tab
<code>dcat:theme</code> / category	topic category or controlled theme	✗ No formal theme vocabulary required (only free-text/GCMD keyword)	✓ Partially addressed via the new <code>topicCategory</code> requirement (ISO topic categories double as a coarse theme mapping); a dedicated DCAT theme vocabulary is not implemented
<code>accessLevel</code>	n/a — POD-specific	✗ Not modeled at all	✗ Removed from scope. Added in v1.1.0, then removed in v1.2.0 — no native ISO/mdJSON element and no mdEditor UI component exist, so requiring it only produced unsatisfiable records. See README Known Limitations for the recommended downstream approach.
<code>bureauCode</code> / <code>programCode</code>	n/a — POD-specific	✗ Not modeled	✗ Removed from scope. Same reasoning and disposition as <code>accessLevel</code> above.
<code>dct:spatial</code> (bounding box)	<code>extent.geographic</code>	✗ Disabled in profiles	✓ Fixed — same fix as ISO geographic extent above
<code>dct:temporal</code>	<code>timePeriod</code>	✓	✓ No change needed
<code>dcat:accrualPeriodicity</code>	<code>maintenance.frequency</code>	⚠ Only exposed (not required) in product profile	⚠ Not addressed — still optional; not blocking for v1.1.0, candidate for a future minor version

DCAT-US / POD Property	Maps from	Baseline status	Status as of v1.1.0
		v1.0.2; absent in v1.0.0/project	
<code>dcat:landingPage</code>	<code>citation.onlineResource</code>	⚠️ Required in project schema, absent as requirement in product schema	⏸️ Not addressed — inconsistency remains, candidate for a future minor version

Baseline conclusion: the biggest blockers to generating valid DCAT-US/POD records were (1) no enforced license/access-rights, (2) distribution format hidden from data stewards, (3) no `accessLevel`/bureau/program codes modeled at all, and (4) publisher not required for projects.

3. File-by-file notes (baseline)

- `ak-proj-schema-v1.0.0.json`: `citation.responsibleParty` had no `contains` constraint for a `publisher` role (unlike the product schema). → **Fixed in v1.1.0.**
- `ak-prod-schema-v1.0.0/1.0.1.json`: Good `publisher` enforcement, but `distribution` items never required `distributionFormat`, and there was no constraints/license requirement gating publication. → **Fixed in v1.1.0.**
- `ak-prod-profile-v1.0.0.json` / `v1.0.2.json`: `extent.geographic.containsData: false` and `distributionFormat` hidden under `transferOption` — hid exactly the two fields DCAT needs most (spatial bbox, format). v1.0.2 did turn on `maintenance.frequency` and `note`, good for accrual periodicity, but not required in schema. → **Extent/format fixed in v1.1.0; accrual periodicity remains optional.**
- `ak-proj-profile-v1.0.0/1.0.1.json`: `extent.geographic.containsData: false` again; also no `constraints` component/nav tab in either version. → **Fixed in v1.1.0** (extent un-hidden, Constraints tab added to nav).
- `ak-data-dict-schema-v1.0.0.json`: No direct ISO or DCAT analog — data dictionaries aren't a first-class DCAT concept. **Left unchanged.** If dictionary content needs to be independently discoverable via DCAT, expose it as a `dcat:Distribution` of the parent product record rather than modifying this schema.

4. Recommended changes — original prioritization and current disposition

High priority (blocks basic ISO/DCAT validity):

1. Re-enable and require geographic extent for spatial resource types → ✅ Done in v1.1.0
2. Require `distributionFormat` → ✅ Done in v1.1.0
3. Add a `constraints` requirement → ✅ Done in v1.1.0
4. Add the missing `publisher` block to the project schema → ✅ Done in v1.1.0

Medium priority (needed for federal DCAT-US/Data.gov submission specifically):

5. `accessLevel` as a required enum → ❌ Added in v1.1.0, **removed in v1.2.0** — no UI component exists in mdEditor; pursue downstream of the tool if Data.gov submission becomes a concrete requirement 6. `bureauCode`/`programCode` → ❌ Same disposition as `accessLevel` 7. Require `maintenance.frequency` (accrual periodicity) consistently → ⏸️ Not done — future minor version

Lower priority (rounds out ISO core conformance):

8. Dataset language and topic category as required fields → ✅ Done in v1.1.0 9. Reference system (CRS) and spatial representation type → ✅ CRS required in schema (⚠️ profile field path unverified); spatial representation type not addressed 10. Require lineage for products → ✅ Done in v1.1.0 (projects out of scope — no lineage tab exists) 11. Record metadata standard name/version → ❌ Added in v1.1.0, **removed in v1.3.0/v1.4.0** — no confirmed mdJSON field path exists; reintroduce only if a valid path is found

5. Remaining work / not carried into v1.1.0

These were identified in the original analysis but intentionally deferred rather than bundled into this release, to keep v1.1.0's breaking changes scoped and reviewable:

- Formal DCAT theme/category vocabulary beyond ISO topic categories
- Consistent `(dcat:landingPage)` (online resource) requirement across product and project schemas
- Required (not just optional) `(accrualPeriodicity)`
- Spatial representation type as a required ISO element
- Custom mdEditor UI components for `(accessLevel)`, `(bureauCode)`, `(programCode)` if Data.gov/POD submission becomes a concrete goal — these fields were removed from the schema in v1.2.0 rather than left as unsatisfiable requirements (see README)
- A confirmed mdJSON field path for metadata standard name/version — removed in v1.3.0/v1.4.0 after no valid path could be located; this is a real ISO element and should be reintroduced once a correct path is confirmed (see README)

6. Practical path forward

Since mdJSON → ISO 19139/19115-3 XML conversion is handled by **mdTranslator** (the engine behind mdEditor), the v1.1.0 schema/profile changes should produce compliant ISO output automatically for the elements now required. For DCAT-US specifically, two paths remain open:

- A custom mdTranslator output template that maps mdJSON → POD JSON, or
- A downstream crosswalk step (several agencies use FGDC's or EPA's published ISO→DCAT-US crosswalks as a starting point) that fills in the POD-only fields `(accessLevel)`, `(bureauCode)`, `(programCode)` that don't exist in mdJSON at all.

See the [README](#) for migration/rollout guidance and open decisions that affect how far this work should go.